

Design a drift path for unmanned surface vehicles using nonlinear Model Predictive Control

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Abstract—With the advancement of unmanned driving technology, Unmanned Surface Vehicles (USVs) have become a significant topic in marine exploration, environmental monitoring, and military applications. USVs are characterized by their efficiency, flexibility, and remote-control capabilities, playing an increasingly important role in surface operations. As the number of USVs increases, the issues of automated charging, maintenance, and management will become substantial time and labor resource burdens. However, the automatic docking of USVs faces numerous technical challenges. Due to the differences in driving mechanisms, inertia, and resistance between vessels and vehicles, the kinematic principles applicable to vehicles are not suitable for small boats, making it impossible to employ parallel parking methods used for vehicles directly. Consequently, using drift docking technology for USVs has emerged as a critically important solution, enabling fully automated management of unmanned fleets, including automatic cruising, docking, and charging, thereby reducing labor and time-related maintenance costs.

This paper uses nonlinear model predictive control and the Gazebo platform as the algorithm and simulation environment for USV drift parking. A dual-propeller USV controller is designed using a simplified kinematic-dynamic bicycle model. The simulation results show that this method can successfully operate on the Gazebo platform and has the possibility of being implemented in an environment where unmanned ships sail in real waters.

Keywords—Unmanned Surface Vehicle (USV), Model Predictive Control (MPC), Twin propeller boat, Gazebo

I. INTRODUCTION

Drifting refers to the significant lateral movement of the vehicle when turning. In this state, the rear tires of the vehicle are locked and have high lateral slip, and the front wheel steering angle is small to complete the turn. While most drivers cannot control the vehicle while drifting, professional racers can control the vehicle and use their skills in drift competitions or rally races. Typically, a racer can initiate a drift by momentarily using the accelerator or brakes to cause the rear tires to suddenly drive or stop. During a drift, they are able to keep the sideslip angle constant and follow the desired path by using counter-steering and throttle inputs. Drift occurs primarily in rally races and dirt tracks where the coefficient of friction between tires is low or

varies greatly, such as on snow or gravel roads. In this case, you can turn faster when using drift [1].

Different from the purpose of drifting for vehicles, small and medium-sized ships use transient drift (operation with non-constant speed) to park parallel to the shore in order to save time and fuel costs when docking at the shore. However, on unmanned ships, there has not yet been any discussion about using drift to dock automatically. USV's drift docking technology relies on ship dynamics, inertia theory and precise automatic driving algorithms [2, 3]. When a ship runs on the water, the hull generates inertia. Drift docking technology accurately controls the docking process by calculating the inertia of the ship, allowing the ship to accurately determine the angle and inertia between the hull and the parking space and adjust the speed in a timely manner to ensure that the hull can smoothly glide into the parking space. Wind and current are important external factors that affect the drifting of unmanned ships to the shore. The USV's autonomous driving system must monitor wind speed, wind direction and water flow rate in real time, and input these data to the control system for calculation [4]. The USV adjusts the course and speed according to these external factors, allowing the ship to use inertia to achieve the goal of parallel docking [5-7]. Therefore, the USV's autonomous driving system needs to use a variety of sensors to sense the surrounding environment, including radar, inertial measurement unit (IMU), visual sensors, and GPS. The data provided by these sensors can help unmanned ships conduct real-time environmental analysis and formulate reasonable path strategies based on the current parking position and space, wind speed, water flow, obstacles and other factors in the actual environment. Therefore, the processing system must have fast and efficient computing capabilities to ensure that the unmanned ship can make timely adjustments according to the situation when approaching the shore.

The control of unmanned vehicles relies on Model Predictive Control (MPC) and a constrained nonlinear optimization algorithm to adjust a series of control inputs to minimize a loss function. The loss function must be calculated based on the difference between the output of the vehicle dynamics model and the reference path. In order to allow USVs to sail on real waters and realize automatic parallel parking through drifting, this study proposes a method that uses nonlinear MPC to

calculate the optimal path for drifting to the shore, and uses Gazebo to simulate and test the possibility of this method.

II. RELATED WORK

Over the past few years, much of the literature's discussion of drift has focused on vehicle control models. There are some studies on drifting and controlling vehicles at the handling limit, such as how professional drivers control cars in drifts [8, 9]. Another way to gain insight into the nature of drift is to analyze the steady-state drift equilibrium. Several studies have shown that steady-state drift can correspond to the vehicle being at an unstable equilibrium point when turning, and using drift at this time is a safer, easier, and more predictable way to turn [10]. Additionally, making it possible to turn on different trajectories while drifting. Using drift in the right situation will help improve the safety of your vehicle. Drifting is a common parking method used on small and medium-sized ships. Quickly completing the landing by drifting can save a lot of time and fuel costs. When the unmanned ship returns to the shore, it still has to be manually towed and stopped, which costs a lot of money.

Unmanned ships with traditional steering gears not only have a large turning radius and take a long time, but also have a complicated structure. In order to navigate complex waters, modern small surface unmanned ships require high flexibility in hull steering, so most of them are driven by twin propellers. This method is a design common in ships and propeller systems. It changes the direction of the ship by adjusting the speed difference of the propellers on both sides. The left and right propellers of the ship generate thrust respectively, and the relationship between the rotational speed difference and the direction of travel is shown in (1).

$$T = k \cdot n^2 \quad (1)$$

where T is the thrust generated by the single-sided propeller (N). K is the thrust coefficient of the propeller and depends on the propeller geometry and water flow characteristics. n is the propeller speed (RPM). These two thrusts are determined by the propeller speed and environmental parameters.

Considering the influence of the propeller's total propulsion force and torque on the ship's traveling direction, the total propulsion force ($F = T_L + T_R$) acts on the longitudinal direction of the ship to make the ship move forward in a straight line. Equation (2) describes that the yaw moment M causes the ship to rotate, thereby changing the course.

$$M = (T_L - T_R) \cdot d \quad (2)$$

where M is the yaw moment (Nm) experienced by the ship. d is the lateral distance (m) between the two propellers. ($T_L - T_R$) is the propeller thrust difference (N).

According to (1), the thrust of the left and right propellers is substituted into (3)

$$T_L = K \cdot n_L^2, \quad T_R = K \cdot n_R^2 \quad (3)$$

Equation (4) describes the thrust difference between the left and right propellers (ΔT)

$$\Delta T = T_L - T_R = K \cdot (n_L^2 - n_R^2) \quad (4)$$

where n_L is the rotational speed of the left propeller. n_R is the rotational speed of the right propeller.

Define the speed difference $\Delta n = (n_L + n_R)$, and we can derive (5) and (6)

$$n_L^2 - n_R^2 = (n_L - n_R) \cdot (n_L + n_R) = \Delta n \cdot (n_L + n_R) \quad (5)$$

$$\Delta T = K \cdot \Delta n \cdot (n_L + n_R) \quad (6)$$

Considering the rotational motion of the ship, the yaw angular velocity (ω) is proportional to the yaw moment (M):

$$\omega = \frac{M}{I_z} \quad (7)$$

where ω is the yaw angular velocity (rad/s), and I_z is the yaw moment of inertia of the ship.

Equation (2) into (4) to get (8):

$$\omega = \frac{\Delta T \cdot d}{I_z} = \frac{K \cdot \Delta n \cdot (n_L + n_R) \cdot d}{I_z} \quad (8)$$

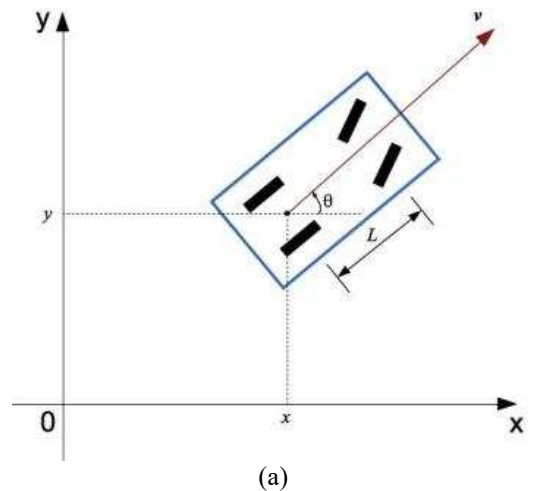
The direction of travel of the ship is determined by the total thrust F and the yaw angular velocity ω . Assuming that the initial course of the ship is zero, the offset angle of the traveling direction θ can be expressed by (9) and (10) as:

$$\theta = \int \omega dt \quad (9)$$

$$\theta = \frac{K \cdot \Delta n \cdot (n_L + n_R) \cdot d}{I_z} \cdot t \quad (10)$$

where t is time, represents the duration of yaw.

The effect of the rotational speed difference of the twin propellers on the traveling direction and yaw angular velocity is applicable to the ideal case of symmetrical propellers and neglecting water flow disturbance. In order to more easily match the ship motion model with the MPC, we use the bicycle model [11, 12] simplified by the Ackerman steering model [11]. As shown in Fig. 1(a), the ship is located in the Global Cartesian coordinate space, where Characterized by its position and heading (x, y, θ), where θ is the heading relative to x -axis. The length of the ship is L , and its speed along the forward direction (θ) is v . The ship moves a certain distance along the forward direction from the starting point (d) (the center of the rear axle in Fig. 1(a)). Fig. 1(b) describes the rotation angle produced by the ship during the movement, where θ can be expressed by (11) is derived.



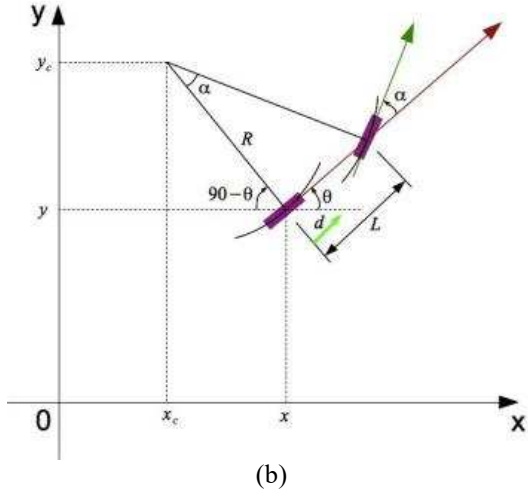


Fig.1 kinematic-dynamic bicycle model [12, 13]

$$\theta = \frac{v}{R} \quad (11)$$

where R is the radius of turn, which can be expressed by (12).

$$R = \frac{L}{\tan \alpha} \quad (12)$$

The slip angle (β) and rotation center coordinates (x_c, y_c) of the hull are given by (13) and (14), respectively.

$$\beta = \int \theta dt = \frac{d}{L} \cdot \tan \alpha = \frac{d}{R} \quad (13)$$

$$x_c = x - R \sin \theta \quad (14)$$

$$y_c = y + R \cos \theta$$

The new course of the ship after turning can be obtained by adding (13) to the original course through β . The updated position and heading are given by (15).

$$x' = x_c + R \sin(\theta + \beta) \quad (15)$$

$$y' = y_c - R \cos(\theta + \beta)$$

$$\theta' = (\theta + \beta) \bmod 2\pi$$

When β and R are close to 0, it can be regarded as a straight-line motion along the heading direction, such as (16).

$$x' = x + d \cos \theta \quad (16)$$

$$y' = y_c - R \cos(\theta + \beta)$$

$$\theta' \approx \theta \bmod 2\pi$$

In the next section, we introduce the use of MPC to infer and transform quadratic programming problems.

III. SYSTEM DESCRIPTION

This paper proposes to use Python and Gazebo as a small unmanned ship to simulate dynamic drift and docking. Python program develops MPC-based kinematic-dynamic bicycle model, course adjustment, speed control and obstacle avoidance technology. Import the developed MPC into Gazebo. The Gazebo platform can be equipped with various sensors, such as images, radars, etc., and adjust and correct the attitude and speed of the hull based on the data returned by these sensors and real-time conditions to achieve smooth and accurate operation. Drifting to the shore. The ship model uses the relationship established between the actuator, external environmental disturbance (wind) and the movement of the hull in the water to

describe the forces acting on the hull through Fossen's kinematic model [14-16]:

$$\dot{\eta} = R(\varphi)v \quad (16)$$

where η is the pose vector including position and yaw angle. $R(\varphi)$ is the horizontal plane rotation matrix φ due to the yaw angle. v are the velocity vectors in pitch, sway and yaw directions respectively. The forces generated by wind, waves, propeller, hull friction and inertia are given by (17).

$$M_{RB}\dot{v} + C_{RB}(v)v + M_A\dot{v}_r + C_A(v_r)v_r + D_{v_r}v_r + D_n(v_r)v_r = \tau_c + \tau_{wi} + \tau_{wa} \quad (17)$$

where M_{RB} notes the rigid body mass matrix. C_A and C_{RB} note matrices describing the Coriolis/centripetal forces, and M_A notes the added mass matrix. $v_r = v - v_c$ notes the relative ship velocity, and $v_c = [uc \ vc \ 0]^T$ is the current velocity. D and $D_n(v_r)$ note linear and nonlinear damping matrices due to the hull moving through water. τ_c , τ_{wi} and τ_{wa} note forces on the ship due to thrusters, wind, and wave effects, respectively.

The path planning problem of drifting to shore can use nonlinear MPC in Python to predict object behavior. Setting the current state of the USV, MPC can use Python's qpsoolvers package to find an optimal control sequence that minimizes cost and meets specified constraints over the entire prediction horizon. A typical general optimization problem for an MPC could look like (18) [17,18]

$$\min_{x,u} J(x, u, x_{ref}) = \sum_{i=0}^{N+1} \|x_i - x_{ref,i}\|_Q^2 + \sum_{i=0}^N \|\Delta u_i\|_R^2 \quad (18)$$

$$\text{subject to } x_{i+1} = f(x_i, u) \quad \forall i \in 0 \dots N \quad x_i = x_{init}, x_i \in \mathcal{X}, u = U$$

where cost function $J(x, u, x_{ref})$ is to be minimized with regard to variables x and u , subject to the model of the process, initial conditions where the first value of the prediction horizon, x_0 , is equal to x_{init} , and the constraints for the variables, $\mathcal{X}$ and U . The difference between state x_i and the reference value $x_{ref,i}$, i for each sample is penalized with a weight matrix Q . Changes in control signals, $\Delta u_j = u_j - u_{j-1}$, are penalized with weight matrix R , to avoid oscillation. The cost function is summed over the prediction horizon N .

Consider a two-dimensional problem where a boat moves in the XY plane. The state of the USV is given by its position (x, y) and its velocity v . The control inputs are the velocity v and the direction a of the USV. The drift is represented by a velocity vector $d(x, y)$, which depends on the position of the USV. The general optimization structure of drift parking, such as (19).

$$\begin{aligned} X &= v \cdot \cos(a) + d_x(x, y) \\ Y &= v \cdot \sin(a) + d_y(x, y) \end{aligned} \quad (19)$$

where d_x and d_y are the components of the drift vector.

The drift function is defined as:

$$d(x, y) = \begin{bmatrix} 3 + 0.2y \cdot (1 - y) \\ 0 \end{bmatrix} \quad (20)$$

Hence, when the USV reaches a specific target location, the speed of the unmanned ship is stationary ($v = 0$). The parking cost function $S_{park}(x_n)$ is defined as a set of target coordinates of the parking space, which is the square error between (X_g, Y_g, φ_g) and the main body coordinates (X_n, Y_n, φ_g) , where n is Sampling points of navigation trajectory [19]:

$$S_{park}(x_n) = (X_g - X_n)^2 + (Y_g - Y_n)^2 + (\varphi_g - \varphi_n)^2 + (v_{xn})^2 + (v_{yn})^2 + (r_n)^2 \quad (21)$$

where zero velocity at the end of the trajectory is desired. The drift parking optimization set the additive cost w to 0.

Before drifting to the shore, the USV first needs to conduct a comprehensive analysis of the surrounding environment through sensors, including evaluation of factors such as avoiding obstacles on the path and the location of the docking space on the shore, wind speed, and currents. Gazebo modeling is used to predict the path of the ship's hull in an environment with or without obstacles. The purpose is to develop a reasonable program for the drifting to shore process and ensure that the ship can approach the shore smoothly. We consider the dynamic ship-stopping operation, in which the USV starts from the origin stationary and reaches the desired coordinate position $(X_g, Y_g, \pi_g) = (10, 5, \pi)$, and the time range is 2 seconds. The drift trajectory returned by the optimization framework is shown in Fig. 3. The blue path is the best path, but there are obstacles detected on this path, so the system selects the second-best path (red) for navigation.

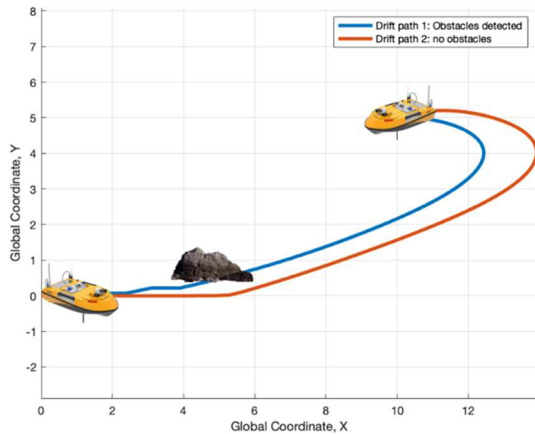


Fig. 3 Nonlinear MPC calculates USV drift docking route in Gazebo

When the USV approaches the shore, the system will adjust according to (21). The USV will enter the drift state by adjusting the speed, keep gliding, and adjust the hull attitude according to environmental factors, as shown in Fig. 4. At this time, the USV's control system will continuously correct the speed and angle to maintain the best path to the shore.

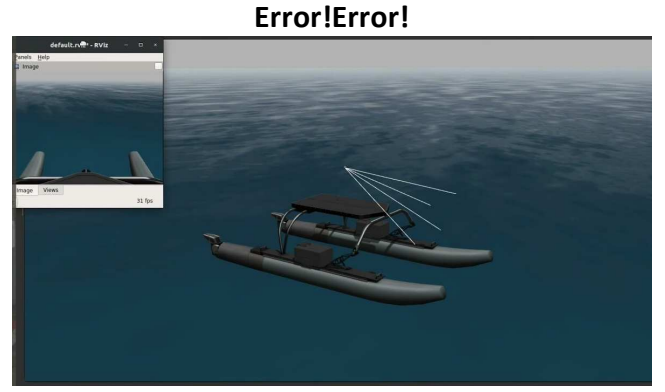


Fig. 4 Gazebo simulates a small unmanned ship for drift parking

When the unmanned ship approaches the dock, the side thrusters will start working to make fine lateral adjustments. The function of the side thruster is to keep the hull at an appropriate distance from the dock and control the hull angle to prevent the hull from deviating from the track. The operation of the side thrusters is adjusted by the autopilot system based on real-time sensor data to ensure that the hull can dock smoothly. In the final stage of approaching the dock, the unmanned ship will use reverse thrusters to slow down to prevent the hull from hitting the shore at too fast a speed. When the hull reaches the predetermined position, the unmanned ship's fixation system will automatically start to securely secure the ship to the dock, ending the entire parking process.

IV. CONCLUSION

In this paper, we propose the use of nonlinear MPC with a kinematic ship model in the Gazebo platform to plan and control USV drift. Nonlinear MPC can realize the steady-state drift of the USV, verifying the robustness of our method in estimating paths under obstacle conditions and variable initial position conditions, and significantly improving the closed-loop performance. USVs often need to perform equipment maintenance and charging on the shore in monitoring and scientific exploration of the marine environment. Drifting docking technology can help USVs dock accurately on the shore, charge automatically, reduce labor maintenance costs, and improve the management efficiency of unmanned fleets. Future work will further verify the possibility of drifting and docking and automated wireless charging of unmanned ships in real sea areas.

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